Methodological Proposal and Preliminary Results

Perturbation-First Evaluation Framework for Automated Short Answer Grading Systems

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1. Research Question

Do Automated Short Answer Grading (ASAG) systems truly understand student response content, or do they rely on superficial cues? Traditional evaluation metrics — accuracy, Quadratic Weighted Kappa — cannot answer this question: they only measure agreement with human scores under normal conditions.

A grader may achieve high accuracy yet be fundamentally brittle: changing a synonym causes a score flip, inserting a negation goes undetected, or stuffing keywords into a wrong answer raises the grade. None of these failures are visible in standard evaluation.

This work proposes a **perturbation-first evaluation framework** that systematically tests grader behaviour under controlled modifications, producing robustness metrics that complement traditional agreement measures.

2. Positioning in the Literature

The methodological framework builds on the CheckList taxonomy (Ribeiro et al., 2020 — ACL Best Paper), which defines three behavioural test types for NLP models: **Invariance**, **Directional Expectation**, and **Minimum Functionality Tests**. This taxonomy has never been systematically applied to ASAG with dedicated metrics.

Existing studies address individual adversarial attack types against ASAG systems: Filighera et al. (2024) demonstrate adjective/adverb gaming attacks reducing grader accuracy by 10–22 percentage points; Ding et al. (2020) show that random-string inputs receive high scores in 30–60% of cases. However, no study proposes a multi-family protocol with complementary metrics for ASAG evaluation. A systematic search across five target journals (*Computers & Education*, *Assessment in Education*, *Language Testing*, *Applied Measurement in Education*, *British Journal of Educational Technology*) returned zero results on adversarial evaluation of ASAG systems.

The construct validity literature provides the theoretical grounding: Ferrara and Qunbar (2022) argue that AI-based scores require evidence of construct-relevant feature use, not just correlation with human ratings. Williamson et al. (2012) explicitly list robustness to gaming as an evaluation criterion. Our perturbation families operationalise these requirements empirically.

The bibliography supporting this work comprises 33 verified entries across four themes: ASAG methods (11), adversarial NLP (10), educational validity (5), and LLM-based assessment (7). Of these, 32 are peer-reviewed; the sole exception is the GPT-4 Technical Report (OpenAI, 2023), which is standard practice. Every entry has been verified against its publisher page or DOI.

3. Datasets

The framework is built and validated on the SemEval 2013 Task 7 Student Response Analysis benchmark (Dzikovska et al., 2013), the most widely used ASAG dataset in the literature.

Corpus	Domain	Questions	Answers	Grading scale
Beetle	Electricity & circuits	42	5,199	5-way (correct, partially correct, contradictory, irrelevant, non-domain)
SciEntsBank	General science	195	10,804	5-way (same)

All labels are normalised to [0.0, 1.0]: correct = 1.0, partially correct = 0.5, all others = 0.0. This enables cross-dataset comparison and continuous metric computation.

Extensibility: The system architecture supports additional datasets through a pluggable loader interface. Integrating a new dataset (e.g., Prof. De Mauro's corpus) requires only a loader module (~50 lines); no changes to perturbation generators, metrics, or the evaluation engine. See Section 8 for details.

4. Methodology

4.1 Dual evaluation protocol

Protocol A — Leave-One-Question-Out (LOQO): The grader is trained on all questions except one and tested on the held-out question. This is repeated for every question. Protocol A measures cross-question generalisation — the real-world scenario where a grader must assess answers to questions it has never seen during training.

Protocol B — Within-Question (80/20 split): For each question, 80% of answers are used for training and 20% for testing. Protocol B measures performance under optimal conditions — when the grader already knows the question.

The comparison between A and B yields the **robustness drop**: the degree to which grader performance degrades on unseen questions. This delta is a direct indicator of deployment

fragility that no traditional metric captures.

4.2 Three perturbation families

Each family tests a distinct dimension of grader behaviour, following the CheckList taxonomy:

Family 1 — Invariance (score must NOT change)

These perturbations modify surface form without altering meaning. If the score changes, the grader is reacting to superficial features.

Type	Description	Example
Synonym substitution	Replace a word with a WordNet synonym	"provides" → "supplies"
Typo insertion	Introduce a realistic character-level error	"because" → "becase"

Family 2 — Sensitivity (score MUST change)

These perturbations alter the semantic content of the answer, making it incorrect. If the score does not change, the grader is not understanding the content.

Type	Description	Example
Negation insertion	Insert "not" / "does not"	"the current flows" → "the current does not flow"
Key concept deletion	Remove an essential term	"The battery provides energy" → "The provides energy"
Semantic contradiction	Replace a term with its antonym	"positive terminal" → "negative terminal"

Family 3 — Gaming (score must NOT increase)

These perturbations attempt to deceive the grader by exploiting surface-level cues. If the score increases, the grader is vulnerable to manipulation.

Type	Description	Example
Rubric keyword echoing	Append keywords from the reference answer	"...connected. separated circuit energy"
Fluent wrong extension	Append a plausible but incorrect sentence	"...Furthermore, batteries produce electrons from the plastic casing."

4.3 Metrics

Metric	Family	What it measures	Formal definition
IVR_{flip}	Invariance	% of cases where score changes when it should not	Pairs where $round(s_{orig}) \neq round(s_{pert})$ / total pairs
$IVR_{absdelta}$	Invariance	Mean magnitude of score change	Mean of $\lvert s_{orig} - s_{pert} \rvert$ over invariance pairs
$SSR_{directional}$	Sensitivity	% of cases where score decreases after semantic alteration	Pairs where $s_{pert} < s_{orig}$ / total pairs
$ASR_{thresholded}$	Gaming	% of failing answers that cross the pass threshold	Pairs where $s_{orig} < 0.5$ and $s_{pert} \geq 0.5$ / total pairs

Both a binary form (IVR_{flip}: "did it change?") and a continuous form ($IVR_{absdelta}$: "by how much?") are computed for invariance, as they capture complementary information.

4.4 Two-gate invariance validation

Invariance perturbations must provably preserve meaning. A two-stage quality gate ensures this:

Gate 1 — SBERT semantic similarity (≥ 0.85): Compares the original and perturbed answer using sentence embeddings (all-MiniLM-L6-v2). Perturbations falling below 0.85 cosine similarity are discarded. The threshold is fixed (not tuneable) to prevent threshold-shopping. Gate 1 applies only to synonym substitutions — typos are meaning-preserving by construction.

Gate 2 — Negation/antonym heuristic: Verifies that no accidental negations or antonyms were introduced. This catches cases that SBERT may miss.

Discarded perturbations are not regenerated. The rejection rate is recorded and reported as a research finding in itself.

4.5 Reproducibility

Every experiment is fully deterministic. Running the same configuration twice produces identical results. This is achieved through per-generator local random seeds, alphabetically sorted WordNet synonyms, JSONL perturbation caching with hash keys, and complete run configuration logging.

5. Grading Models

The framework compares four grading approaches, from simple to state-of-the-art:

Model	Type	Training	Status
HybridGrader	Linguistic features + SBERT embeddings + logistic regression	Supervised on dataset	Done
DeBERTa-v3-base	Pre-trained transformer (He et al., 2023)	Fine-tuned on dataset	Next
ModernBERT-base	Pre-trained transformer (Answer.AI)	Fine-tuned on dataset	Planned
GPT-4o	Large Language Model (OpenAI, 2023)	Zero-shot with rubric prompt	Planned

The multi-model comparison is essential: if even sophisticated models exhibit fragility under perturbation, the finding is substantially stronger than a single-model analysis.

6. Preliminary Results

A complete experimental run has been executed on the HybridGrader with the Beetle corpus (42 questions, 5,199 answers, ~41,000 perturbations).

6.1 Perturbation generation

Statistic	Value
Answers processed	5,199
Perturbations generated	41,378
Mean perturbations per answer	8.0 (out of 10 theoretical; some rejected by gates)
All 7 perturbation types present	Yes
Gate 1 rejection rate (synonyms)	40.3%
Gate 2 rejection rate	0%

The 40.3% Gate 1 rejection rate is itself a finding: nearly half of WordNet synonym substitutions alter sentence-level semantics sufficiently to fall below the SBERT similarity threshold. This

demonstrates that naive synonym substitution is not a reliable meaning-preserving transformation without validation.

6.2 Main results (HybridGrader × Beetle)

Metric	What it measures	Protocol B (known questions)	Protocol A (unseen questions)	Robustness Drop
IVR_{flip}	Synonym instability	34.1%	53.9%	+19.8 pp
$IVR_{absdelta}$	Instability magnitude	0.233	0.402	+16.9 pp
$SSR_{directional}$	Negation sensitivity	16.6%	32.3%	+15.7 pp
$ASR_{thresholded}$	Gaming vulnerability	19.2%	18.0%	−1.2 pp

pp = percentage points. Runtime: ~83 min on Apple M1.

6.3 Interpretation

IVR_{flip} = 53.9% (Protocol A): On unseen questions, the grader changes its score in over half of the cases after a simple synonym substitution. In a real deployment, the same concept expressed with different words would receive a different grade 54% of the time. This level of instability would be unacceptable for educational use.

$SSR_{directional}$ = 32.3% (Protocol A): The grader detects negation and lowers the score in only 32% of cases. In the remaining 68%, an answer made semantically incorrect by inserting "not" or replacing a term with its antonym is not penalised. The grader is reacting to surface patterns, not semantic content.

$ASR_{thresholded}$ = 18.0% (Protocol A): 18% of gaming attempts succeed in converting a failing answer to a passing one. This is relatively contained but still non-trivial.

The robustness drop is the most significant finding. The systematic delta between Protocol A and Protocol B — +19.8 pp (invariance), +16.9 pp (magnitude), +15.7 pp (sensitivity) — confirms the central hypothesis: robustness performance degrades measurably when the grader faces unseen questions. The consistency across three independent metrics strengthens the validity of the result.

Gaming vulnerability is the exception (−1.2 pp), remaining stable across protocols. This suggests that keyword-stuffing vulnerability is question-independent — a plausible finding, since gaming exploits general lexical patterns rather than question-specific knowledge.

7. Experimental Plan

Phase	Content	Dependencies
Phase 4 Next	Fine-tune DeBERTa-v3-base and ModernBERT-base under both LOQO and within-question protocols, with 3 random seeds per model. Produces the same metric table as the HybridGrader baseline for direct comparison.	GPU access
Phase 5 Planned	Evaluate GPT-4o (and at least one other LLM) via API with rubric-conditioned prompts. Includes API call caching, cost estimation, and LLM consistency rate measurement across repeated runs.	API budget
Phase 6 Planned	Full statistical analysis: Wilcoxon signed-rank with Bonferroni correction, Cliff's delta for effect size, Kendall's tau for metric agreement. Publication-ready outputs: LaTeX tables, perturbation × model heatmap, dumbbell plots (clean vs. perturbed), radar chart per model.	—

Expected output table for the paper:

	IVR_{flip}	$IVR_{absdelta}$	SSR_{dir}	ASR_{thresh}
HybridGrader (A)	53.9%	0.402	32.3%	18.0%
HybridGrader (B)	34.1%	0.233	16.6%	19.2%
DeBERTa-v3 (A)	*pending*	*pending*	*pending*	*pending*
DeBERTa-v3 (B)	*pending*	*pending*	*pending*	*pending*
ModernBERT (A)	*pending*	*pending*	*pending*	*pending*
ModernBERT (B)	*pending*	*pending*	*pending*	*pending*
GPT-4o (A)	*pending*	*pending*	*pending*	*pending*
GPT-4o (B)	*pending*	*pending*	*pending*	*pending*

8. Integration of Additional Datasets

The framework architecture was designed from the outset to support additional datasets without modification to the perturbation generators, metrics, or evaluation engine. Integrating a

new dataset requires only a loader module that maps the raw data to the framework's canonical schema (QuestionRecord, AnswerRecord).

To integrate a new corpus, the following information is needed:

- File format (CSV, JSON, Excel)
- Available fields (question text, student answer, reference answer, score)
- Grading scale used
- Rubric or marking guide (if available — used for LLM grader prompts)

Once the format is defined, the loader can be implemented in a few hours and all experiments already conducted on Beetle/SciEntsBank can be replicated automatically on the new dataset, producing directly comparable results.

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Full bibliography (33 entries) available in the project repository.